The Cone Generated by Positive Semidefinite Mean Polynomials

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Abstract

Certain classes of (homogeneous) polynomials, such as Positive Semidefinite (PSD) Diagonal minus Tail (D-T) forms and those derived from the Arithmetic-Geometric Mean inequality, like Sums Of Nonnegative Circuit (SONC) polynomials, can be identified with nonnegative mean polynomials. This paper studies PSD forms that are weighted power means, denoted by $M_{q,p}(Y, w)$, and investigates the cone of nonnegative mean real n-variate forms of degree $2d$, denoted $M_{n,2d}(X, \alpha)$, and its relation to other important cones: the cone of all PSD forms, the cone of Sums of Squares (SOS), and the cone of SONC polynomials. A key object of study is the sum of the SOS and SONC cones, known as the SOSONC cone.

1 Introduction

Certain classes of (homogeneous) polynomials, such as Positive Semidefinite (PSD) Diagonal minus Tail (D-T) forms and those derived from the Arithmetic-Geometric Mean inequality, like Sums Of Nonnegative Circuit (SONC) polynomials, can be identified with nonnegative mean polynomials.

In this paper, we study PSD forms that are weighted power means, denoted $M_{q,p}(Y, w)$, where $Y = (Y_1, \ldots, Y_m)$ is an m-tuple of symbols and $w = (w_1, \ldots, w_m) \in \mathbb{R}_+^m$ is a vector of positive weights. In particular, we investigate the cone of nonnegative mean real n-variate forms of degree $2d$, denoted $M_{n,2d}(X, \alpha)$, where $X = (X_1, \ldots, X_n)$ and $\alpha \in \mathbb{N}_0^n$.

We study the relationships between the following cones:

- $\mathcal{P}_{n,2d}$: the cone of all positive semidefinite (PSD) forms in n variables of degree $2d$; our cone is a subset: $\mathcal{M}_{n,2d} \subseteq \mathcal{P}_{n,2d}$.
- $\Sigma_{n,2d}$: the cone of sums of squares (SOS) polynomials.
- $\mathcal{C}_{n,2d}$: the cone of sums of nonnegative circuit (SONC) polynomials.

A key object of study is the sum of the SOS and SONC cones, known as the *SOSONC cone*, defined by M. Schick:

$$SOSONC = (\Sigma + \mathcal{C})_{n,2d}$$

In §2 (Preliminaries on mean polynomials), we introduce the preliminary concepts of weighted power means, denoted $M_{q,p}(Y, w)$, $q, p \in \mathbb{N}_0$, $Y = (Y_1, \ldots, Y_m)$, $w = (w_1, \ldots, w_m) \in \mathbb{R}_+^m$, and the fundamental tool of Jensen's inequality. We prove (see Proposition 1.1) that if $p < q$, then $M_{q,p}(Y, w)$ is PSD. We identify certain classes of nonnegative polynomials as instances of $M_{q,p}(Y, w)$ for appropriate choices of q, p, Y, and w.

In §3 (Nonnegative Circuit Polynomials), we re-prove [2, Corolary 2.2] and [6, Theorem 2.4] using mean polynomials, giving necessary and sufficient conditions for nonnegativity of D-T forms and circuit polynomials.

In §4 (The cones $M_{2d,0}(\underline{X}, \alpha)$ and $M_{1,0}(Y, \lambda)$), we prove Proposition 3.1: the forms $M_{2d,0}(X, \alpha)$ are PSD D-T for any $d \in \mathbb{N}$ and any $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = 2d$. The cone generated by these forms is the cone of sums of PSD D-T forms.

In Theorem 3.1, we prove that the SONC cone, $\mathcal{C}_{n,2d}$, is the cone generated by the forms $M_{1,0}(Y, \lambda)$.

In §5, we study $M_{2d,p}(X, \alpha)$ with $|\alpha| = 2d$, $p|2d$ in connection to SOBS and SOS. We prove in Proposition 4.1 that $M_{2d,0}(X, \alpha)$ is SOBS and PSD D-T, $|\alpha| = 2d$. In Theorem 4.1, we continue investigating $M_{2d,p}(X, \alpha)$ with $d = 1, 2, 3$, $p|2d$ and prove that they are SOS for all $n \in \mathbb{N}$ (by reduction to Hilbert's case and induction on the number of variables).

In §6 (General Case: Beyond the SOSONC cone), the ultimate goal is to move towards a comprehensive understanding of the general form $M_{q,p}(Y, w)$. In particular, we examine the Choi-Lam form Q and the second Robinson form $\hat{R}$ (see Section 5.1).

In [9], it is observed that Q is SONC [9, page 31], but is not SOSONC [9, Lemma 4.2.5, page 107]. It was further observed by Reznick [10] that $\hat{R}$ can be obtained from Q by a linear change of variables. In Section 5.2, we establish moreover that

$$Q(x, y, z, w) = 4M_{1,0}((x^4, y^4, z^4, w^4), (1, 1, 1, 1))$$

and accordingly

$$\hat{R}(x, y, z, w) = 8M_{1,0}(G, (1, 1, 1, 1))$$

where $G = \frac{1}{2}(x+w-y-z, y+w-x-z, z+w-x-y, w-(x+y+z))$. Mehdi to check the computation. Therefore $\hat{R}$ is a separating form for the cone of mean polynomials and the SOSONC cone. In §7 (Positivestellensatz and relation the the cone of $2d$ powers).

In §**??** (Application to Polynomial Optimization): Mehdi: to check the algorithmic aspect of means

2 Preliminaries on Mean Polynomials

Let $\varphi(x) = x^{\frac{2q}{p}}$, where $q > p$. The second derivative of φ is

$$\varphi''(x) = \frac{q}{p}\left(\frac{q}{p} - 1\right) x^{\frac{2q}{p}-2}$$

Since $q > p$, the coefficient $\frac{q}{p}\left(\frac{q}{p} - 1\right)$ is always positive, and if q is even then φ'' is always positive, so φ is convex. By Jensen's inequality,

$$\varphi\left(\frac{\sum w_i x_i}{\sum w_i}\right) \leq \frac{\sum w_i \varphi(x_i)}{\sum w_i}$$

where $w_i \in \mathbb{R}_+$. Replacing x_i with x_i^p, we get

$$\varphi\left(\frac{\sum w_i x_i^p}{\sum w_i}\right) \leq \frac{\sum w_i \varphi(x_i^p)}{\sum w_i}$$

and hence

$$\left(\frac{\sum w_i x_i^p}{\sum w_i}\right)^{\frac{q}{p}} \leq \frac{\sum w_i x_i^q}{\sum w_i}$$

or equivalently $M_p(x, w) \leq M_q(x, w)$. Raising both sides to the power $c = \operatorname{lcm}(q, p)$, we conclude that the function

$$\mathcal{M}_{q,p} = M_q^c(x, w) - M_p^c(x, w)$$

is always a nonnegative polynomial function in $x = (x_1, \ldots, x_n)$.

Basic properties of M_p

- $\lim_{p \to -\infty} M_p(\mathbf{x}, w) = \min(x_1, \ldots, x_n)$.
- $\lim_{p \to \infty} M_p(\mathbf{x}, w) = \max(x_1, \ldots, x_n)$.
- $\lim_{p \to 0} M_p(\mathbf{x}, w) = \left(\prod_{i=1}^{n} x_i^{w_i}\right)^{1/\sum_{i=1}^{n} w_i}$.
- If $p < q$, then $M_p(\mathbf{x}, w) \leq M_q(\mathbf{x}, w)$. Equality holds when $x_1 = x_2 = \cdots = x_n$.

3 Nonnegative Diagonal minus Tail and Circuit Polynomials

Theorem 1. *Let the function $f(x)$ be defined as*

$$f(x) = \sum_{i=1}^{n} \beta_i x_i^{2d} - \mu x^{\alpha}$$

where $x = (x_1, \ldots, x_n)$ with $x_i \geq 0$, $\beta_i > 0$, and $\alpha = (\alpha_1, \ldots, \alpha_n)$ with $\alpha_i > 0$. The monomial x^{α} is defined as $x^{\alpha} = \prod_{i=1}^{n} x_i^{\alpha_i}$. We assume the degree is conserved, i.e., $\sum_{i=1}^{n} \alpha_i = 2d$.

The function $f(x)$ is positive semidefinite (PSD), meaning $f(x) \geq 0$ for all $x \in \mathbb{R}_+^n$, if and only if the following condition holds:

$$\mu^{2d} \prod_{i=1}^{n} \alpha_i^{\alpha_i} \leq (2d)^{2d} \prod_{i=1}^{n} \beta_i^{\alpha_i}$$

Proof. The proof consists of two parts for the "if and only if" statement.

($\Leftarrow$) Assume the inequality above holds. We want to show that $f(x) \geq 0$. This is equivalent to showing

$$\sum_{i=1}^{n} \beta_i x_i^{2d} \geq \mu x^{\alpha} \quad \forall x \in \mathbb{R}_+^n$$

The proof relies on the weighted Arithmetic Mean-Geometric Mean (AM-GM) inequality. Applying this to the terms $y_i = \frac{\alpha_i}{\beta_i} x_i^{2d}$ with weights $w_i = \alpha_i$ (where $\sum w_i = 2d$), we get

$$\frac{\sum_{i=1}^{n} \beta_i x_i^{2d}}{2d} \geq \left(\prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\alpha_i} \right)^{1/2d} x^{2d}$$

which implies

$$\sum_{i=1}^{n} \beta_i x_i^{2d} \geq 2d \left(\prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\alpha_i} \right)^{1/2d} x^{\alpha}$$

From our initial assumption, we can take the $2d$-th root and rearrange to find

$$\mu \leq 2d \left(\prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\alpha_i} \right)^{1/2d}$$

Combining these results, we chain the inequalities

$$\mu x^{\alpha} \leq 2d \left(\prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i} \right)^{\alpha_i} \right)^{1/2d} x^{\alpha} \leq \sum_{i=1}^{n} \beta_i x_i^{2d}$$

Thus, we have shown $\sum_{i=1}^{n} \beta_i x_i^{2d} \geq \mu x^{\alpha}$, which holds for $x_i > 0$. If any $x_i = 0$, then $x^{\alpha} = 0$ and the inequality becomes $\sum \beta_i x_i^{2d} \geq 0$, which is trivially true.

($\Rightarrow$) Now, we prove the other direction by contradiction. Assume that $f(x)$ is PSD, but the inequality above does *not* hold. This means

$$\mu^{2d} \prod_{i=1}^{n} \alpha_i^{\alpha_i} > (2d)^{2d} \prod_{i=1}^{n} \beta_i^{\alpha_i}$$

Taking the $2d$-th root and rearranging gives

$$\mu > 2d \left(\prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i} \right)^{\alpha_i} \right)^{1/2d}$$

Since we assumed $f(x) \geq 0$ for all $x \in \mathbb{R}_+^n$, we can show a contradiction by constructing a specific vector x for which $f(x) < 0$. We choose the vector x that corresponds to the equality condition of the AM-GM inequality used above. This occurs when all the terms $y_i = \frac{\alpha_i}{\beta_i} x_i^{2d}$ are equal. Let $x_i = \left(\frac{\beta_i}{\alpha_i}\right)^{1/2d}$.

Now we evaluate $f(x)$ for this specific x:

- The sum term is

$$\sum_{i=1}^{n} \beta_i x_i^{2d} = \sum_{i=1}^{n} \beta_i \left(\frac{\beta_i}{\alpha_i} \right) = \sum_{i=1}^{n} \alpha_i = 2d$$

- The monomial term is

$$x^\alpha = \prod_{i=1}^{n} x_i^{\alpha_i} = \prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i} \right)^{\alpha_i/2d} = \left(\prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i} \right)^{\alpha_i} \right)^{1/2d}$$

Substituting these into $f(x)$:

$$f(x) = 2d - \mu \left(\prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\alpha_i} \right)^{1/2d}$$

This implies that $2d - \mu \left(\prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\alpha_i} \right)^{1/2d} < 0$. So, for our chosen x, we have found that $f(x) < 0$. This contradicts the premise that $f(x)$ is PSD. Therefore, our assumption that the inequality does not hold must be false. Both directions of the proof are complete. $\square$ $\square$

Theorem 2. *Let $\alpha(0), \alpha(1), \ldots, \alpha(r) \in \mathbb{N}^n$ be exponent vectors. Let the exponent vector $\beta \in \mathbb{N}^n$ be a convex combination of the $\alpha(j)$ vectors, such that $\beta = \sum_{j=0}^{r} \lambda_j \alpha(j)$, with $\lambda_j > 0$ and $\sum_{j=0}^{r} \lambda_j = 1$. Consider the polynomial $f(X)$ with real coefficients $f_\alpha > 0$ defined as*

$$f(X) = \sum_{j=0}^{r} f_{\alpha(j)} X^{\alpha(j)} - f_\beta X^\beta$$

where $X = (X_1, \ldots, X_n)$.

The polynomial $f(X)$ *is positive semidefinite (PSD) for all* $X \in \mathbb{R}_+^n$ *if and only if*

$$|f_\beta| \leq \prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j}$$

Proof Sketch. The proof of the ($\Leftarrow$) direction relies on the weighted Arithmetic Mean-Geometric Mean (AM-GM) inequality. Define terms Y_j and use the weights λ_j:

$$Y_j = \frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)}, \quad j = 0, \ldots, r$$

The weighted AM-GM inequality states that the arithmetic mean is greater than or equal to the geometric mean:

$$\sum_{j=0}^{r} \lambda_j Y_j \geq \prod_{j=0}^{r} Y_j^{\lambda_j}$$

- **Arithmetic Mean (LHS):**

$$\sum_{j=0}^{r} \lambda_j Y_j = \sum_{j=0}^{r} f_{\alpha(j)} X^{\alpha(j)}$$

- **Geometric Mean (RHS):**

$$\prod_{j=0}^{r} Y_j^{\lambda_j} = \prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j} X^\beta$$

Combining these, the AM-GM inequality implies

$$\sum_{j=0}^{r} f_{\alpha(j)} X^{\alpha(j)} \geq \prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j} X^\beta$$

If we assume the theorem's condition holds ($f_\beta \leq \prod_{j=0}^{r} (\frac{f_{\alpha(j)}}{\lambda_j})^{\lambda_j}$), then it follows that

$$\sum_{j=0}^{r} f_{\alpha(j)} X^{\alpha(j)} \geq f_\beta X^\beta \implies f(X) \geq 0$$

This proves one direction of the theorem. $\square$

Application to D-T Forms

Problem. Find the condition under which the function $f(x)$ is positive semidefinite for $x \in \mathbb{R}_+^n$:

$$f(x) = \sum_{i=1}^{n} \beta_i x_i^{2d} - \mu x^\alpha$$

where $\beta_i, \mu > 0$ and the exponents satisfy $\sum_{i=1}^n \alpha_i = 2d$.

Solution via Theorem 2.2.

We can map this problem to the general theorem's framework:

- The "arithmetic mean" part is $\sum_{i=1}^n \beta_i x_i^{2d}$. The corresponding exponent vectors are $\alpha(i) = 2de_i$, where e_i is the i-th standard basis vector in $\mathbb{R}^n$. The coefficients are $f_{\alpha(i)} = \beta_i$.
- The "geometric mean" part is μx^α. The exponent vector is $\beta = (\alpha_1, \alpha_2, \ldots, \alpha_n)$. The coefficient is $f_\beta = \mu$.
- We must express β as a convex combination of the $\alpha(i)$ vectors: $\beta = \sum_{i=1}^n \lambda_i \alpha(i)$.

This gives

$$(\alpha_1, \ldots, \alpha_n) = \sum_{i=1}^n \lambda_i (0, \ldots, 2d, \ldots, 0) = (2d\lambda_1, \ldots, 2d\lambda_n)$$

which implies that the weights must be $\lambda_i = \frac{\alpha_i}{2d}$. The condition $\sum \lambda_i = 1$ becomes $\sum \frac{\alpha_i}{2d} = 1$, which is exactly the given condition $\sum \alpha_i = 2d$.

Now, we apply the inequality from the general theorem:

$$f_\beta \leq \prod_{i=1}^n (\frac{f_{\alpha(i)}}{\lambda_i})^{\lambda_i}.$$

Substituting our specific terms:

$$\mu \leq \prod_{i=1}^n (\frac{\beta_i}{\alpha_i/2d})^{\alpha_i/2d} = \prod_{i=1}^n (\frac{2d \cdot \beta_i}{\alpha_i})^{\alpha_i/2d}$$

To see that this is the same condition, we can raise both sides to the power of $2d$:

$$\mu^{2d} \leq (\prod_{i=1}^n (\frac{2d \cdot \beta_i}{\alpha_i})^{\alpha_i/2d})^{2d} = \prod_{i=1}^n (\frac{2d \cdot \beta_i}{\alpha_i})^{\alpha_i},$$

Since $\sum \alpha_i = 2d$, this simplifies to:

$$\mu^{2d} \leq \frac{(2d)^{2d} \prod \beta_i^{\alpha_i}}{\prod \alpha_i^{\alpha_i}}.$$

Thus

$$\mu^{2d} \prod \alpha_i^{\alpha_i} \leq (2d)^{2d} \prod \beta_i^{\alpha_i}. \square$$

4 Investigation of $M_{2d,p}(\underline{X}, \alpha)$ and Connection to SOBS and SOS

Proposition 1. *$M_{2d,0}(X,\alpha)$ is PSD D-T and SOBS.*

Proof. See [2]. (A simpler proof may be possible.) □

Theorem 3. *$M_{2d,p}(X,\alpha)$ with $d = 1,2,3$, $p|2d$ is SOS for all $n \in \mathbb{N}$ and $\alpha \in \mathbb{N}^n$ with $|\alpha| = 2d$.*

Proof. **Case $d = 1$:** $M_{2,0}$ and $M_{2,1}$ are PSD quadratic forms and by Hilbert's theorem, they are SOS.

Case $d = 2$: For $p = 0$, these are PSD and SOBS by Proposition 4.1. For $p = 1$, we show that $M_{4,1}(X,\alpha)$ is a sum of squares (SOS).

For any number of variables n and any nonnegative weights $\alpha = (\alpha_1, \ldots, \alpha_n)$ such that $|\alpha| = 4$, the form

$$M_{4,1}(X,\alpha) = \frac{\sum_{i=1}^n \alpha_i X_i^4}{4} - \left(\frac{\sum_{i=1}^n \alpha_i X_i}{4}\right)^4$$

is SOS. We examine each case by the number of variables n:

1. **Case $n = 2$ (Binary Forms):** For $n = 2$, $M_{4,1}$ is a PSD binary quartic form. By Hilbert's Theorem (1888), any nonnegative polynomial in two variables is a sum of squares. Therefore, $M_{4,1}$ is SOS for $n = 2$.

2. **Case $n = 3$ (Ternary Quartics):** For $n = 3$, $M_{4,1}$ is a PSD ternary quartic. Again, by Hilbert's Theorem, any nonnegative ternary quartic is a sum of squares. Therefore, $M_{4,1}$ is SOS for $n = 3$.

3. **Case $n = 4$:**

   - For uniform weights, $\alpha = (1,1,1,1)$:

     $$M_{4,1}(X,(1,1,1,1)) = \frac{\sum_{i=1}^4 X_i^4}{4} - \left(\frac{\sum_{i=1}^4 X_i}{4}\right)^4$$

     This specific form is known to be a sum of squares, verified computationally by semidefinite programming (SDP).

   - For other α with $|\alpha| = 4$ (e.g., $(2,1,1,0)$, $(3,1,0,0)$, $(4,0,0,0)$), all involve at most three non-zero entries, so they reduce to the $n = 3$ case and are SOS.

4. **Case $n \geq 5$:** For any $\alpha = (\alpha_1, \ldots, \alpha_n)$ with $|\alpha| = 4$, there are at most four non-zero entries. Thus, $M_{4,1}(X,\alpha)$ only involves variables corresponding to non-zero weights, reducing to a case with at most 4 variables ($n \leq 4$). Since all cases for $n \leq 4$ are SOS, the proof is complete for $p = 1$.

5. For $p = 2$, $M_{4,2}(X,\alpha)$ is SOS. For any n and α with $|\alpha| = 4$,

   $$M_{4,2}(X,\alpha) = \frac{\sum_{i=1}^n \alpha_i X_i^4}{4} - \left(\frac{\sum_{i=1}^n \alpha_i X_i^2}{4}\right)^2$$

- For $n = 2$ and $n = 3$, by Hilbert's Theorem, these are SOS.
- For $n = 4$, with $\alpha = (1, 1, 1, 1)$:

$$M_{4,2}(X, (1, 1, 1, 1)) = \frac{\sum_{i=1}^{4} X_i^4}{4} - \left(\frac{\sum_{i=1}^{4} X_i^2}{4}\right)^2$$

  This form is nonnegative and computation shows it is SOS.
- For other α with $|\alpha| = 4$, all involve at most three non-zero entries, so they reduce to the $n < 4$ case.
- For $n \geq 5$, any α has at most four non-zero entries, so the problem reduces to $n \leq 4$.

Case $d = 3$: Consider the family of forms

$$M_{6,p}(X, \alpha)$$

where $p \in \{0, 1, 2, 3\}$, $|\alpha| = 6$. We examine by the number of variables n:

1. For $n = 2$ (Binary Forms): By Hilbert's Theorem, all $M_{6,p}$ are SOS for $n = 2$.
2. For $n = 3$ (Ternary Forms):
   - For $p = 0$, $M_{6,0}$ is known to be SOS.
   - For $p = 1$, analyze

$$M_{6,1}(X, \alpha) = \frac{\sum_{i=1}^{3} \alpha_i X_i^6}{6} - \left(\frac{\sum_{i=1}^{3} \alpha_i X_i}{6}\right)^6$$

     Possible partitions of 6 into 3 parts for $\alpha \in \mathbb{N}^3$ are $(2, 2, 2)$, $(3, 2, 1)$, $(4, 1, 1)$, $(5, 1, 0)$, $(6, 0, 0)$.
   - For $\alpha = (2, 2, 2)$: $M_{6,1}(X, (2, 2, 2))$ is known to be SOS.
   - For $\alpha = (3, 2, 1)$ and $(4, 1, 1)$: These can be reduced to $(2, 2, 2)$ by change of variables, so are SOS.
3. For $n = 4$: Possible partitions of 6 into 4 parts for α are $(2, 2, 1, 1)$, $(3, 1, 1, 1)$, $(4, 1, 1, 0)$, $(4, 2, 0, 0)$, $(5, 1, 0, 0)$, $(3, 2, 1, 0)$, $(6, 0, 0, 0)$. Cases with zeros reduce to $n \leq 3$.
   - For $\alpha = (2, 2, 1, 1)$ and $(3, 1, 1, 1)$: These are known nonnegative forms and SDP decomposition shows they are SOS.
4. For $n = 5$ and $n = 6$: Partitions with zeros reduce to fewer variables. For $n = 6$, only $(1, 1, 1, 1, 1, 1)$ is without zeros.

This completes the proof for Theorem 4.1. □

Remark 1. *1. It is of interest to investigate if Theorem 4.1 generalizes to any d.*

2. It would be interesting to know if these forms are in the SONC cone.

5 Beyond the SOSONC Cone

5.1 Specific Examples to Analyze

- $Q(X,Y,Z,W) = X^4 + Y^4 + Z^4 + W^4 - 4XYZW$

Similar as the Motzkin form, one can easily verify that both Choi Lam forms Q, S, and the modified Motzkin M are indeed PSD circuits (page 31, M. Schick thesis).

$$R(X,Y,Z) := X^6+Y^6+Z^6-(X^4Y^2+X^4Z^2+X^2Y^4+X^2Z^4+Y^4Z^2+Y^2Z^4)+3X^2Y^2Z^2$$

$$\hat{R}(X,Y,Z,W) := X^2(X-W)^2+Y^2(Y-W)^2+Z^2(Z-W)^2+2XYZ(X+Y+Z-2W)$$

$$P(X,Y,Z) := 200\left[(X^3-4XZ^2)^2+(Y^3-4YZ^2)^2\right]+(Y^2-X^2)X(X+2Z)\left[X(X-2Z)+2(Y^2-4Z^2)\right]$$

Remarkably, there is the following relation between Q and R [10, see page 5]:

$$\hat{R}(X-W, Y-W, Z-W, X+Y+Z-W) = 2Q(X,Y,Z,W)$$

Since the PSD and SOS cones are closed under linear transformations, the identity above shows that R is PSD (SOS) if and only if Q is PSD (SOS). Now, using the Newton polytope method one can prove that $Q \in P^h_{4,4} \setminus \Sigma^h_{4,4}$. In this sense, the Robinson form R admits a proof of $\hat{R} \in P^h_{4,4} \setminus \Sigma^h_{4,4}$ following the Newton polytope method. (page 38)

5.2 Separating Forms via Linear Transformation of the Variables

6 Positivestellensatz and relation to the cone of $2d$ powers

7 Application to polynomial optimization

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